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Inside the Black Box: A Deep Dive Into Large Language Models

Introduction session, 17.04.2026, 12:00-14:00

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Agenda

1. Introduction from me
2. Introduction from you
3. Time plan
4. Intro to the class
5. Requirements
6. LLMs 101
7. Wrap-Up & Questions

Areas of Interest

- **Language acquisition in the era of LLMs**: how children learn language compared to machines
- **LLMs through the lens of human cognition**: what they reveal (and fail to reveal) about how we process language
- **Meaning and meaning representation**: contrasting semantic representations in humans vs. LLMs
- **LLMs in linguistics**: tasks, theories, language modeling
- **NLP and Corpus Linguistics**: using methods from both for text analysis

Summer School: Teaching

- Co-organized a workshop at *Large Language Models for Digital Humanities Research* (Cologne, 2025)
- Session: **“How Large Language Models Represent Meaning: A Cognitive and Computational Perspective”**
- Covered topics: semantics, distributional semantics, word embeddings, semantic processing in humans, word vectors models, obtaining word embeddings in a neural network

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Introduction Round

- Name
- Language background (first language, spoken languages)
- Study program
- Bachelor’s or Master’s
- Why did you choose to join the class?
- What would you like to learn during the class?

Time Plan

- This is a **Blockseminar**:

  1. **Introduction (17.04.2026, 12:00-14:00)**
  2. Introduction to NLP and (L)LMs (24.04.2026, 12:00-16:00)
  3. How Large Language Models Represent Meaning (08.05, 12:00-16:00) **(ROOM CHANGE)**
  4. Presentations + The Hidden Labor, Research Crisis & Ethical Costs (22.05.2026, 12:00-16:00)
  5. Presentations + Transparency & Open Source (practical part) (12.06, 12:00-16:00)
  6. Presentations + Hands on Research (19.06, 12:00-15:00)

Ground Rules

- Call me Hanna
- Feedback, questions, suggestions what else you would like to learn
- There are no stupid questions
- If something is not clear, always ask to repeat
- Discussions
- The class is in English - all the class, including discussions in groups and presentations

Prerequisites

- Google account for the Google Collab Jupiter notebook
- Bring a laptop

Requirements: Presentation and Presentation Chair

- **Presentation:**
  - You present a chosen paper from the list.
- **Discussion chair:**
  - You prepare 2-3 questions to someone else's presentation.

15 min for presentation, 10 min for questions and discussion

- Active participation

Paper Presentation

1. What the project is about
2. Research gap: why the authors decided to do this project
3. Grounding in literature
4. Methods
5. Results
6. Discussing the results
7. Outlook, limitations, future directions

Presentation Chair

- Introduce the speaker in 1 minute
- Keep time
- Ask the first 2–3 questions
- Moderate audience discussion
- Maybe give a short critical response

Presentations

- If the paper you wanted to present is already taken, you can:
  1. choose another paper from the list
  2. send me your suggestion of a paper

How to Read a Paper

- Step 1: Abstract, Introduction, Conclusions
- Step 2: Look at the figures and tables
- Step 3: Define the research gap and research questions
- Step 4: dive deeper into methodology

Course Materials and Resources

- All the materials available in Github repo:

https://github.com/hannawoloszyn/LLMs-class-ss26

Presentations: Task for Now

- Go to Github repo
- Look through the paper list
- Choose a paper you’d like to present and a paper you’d like to chair
- Put your name in the table for the presentation and presentation chair

Questions so far?

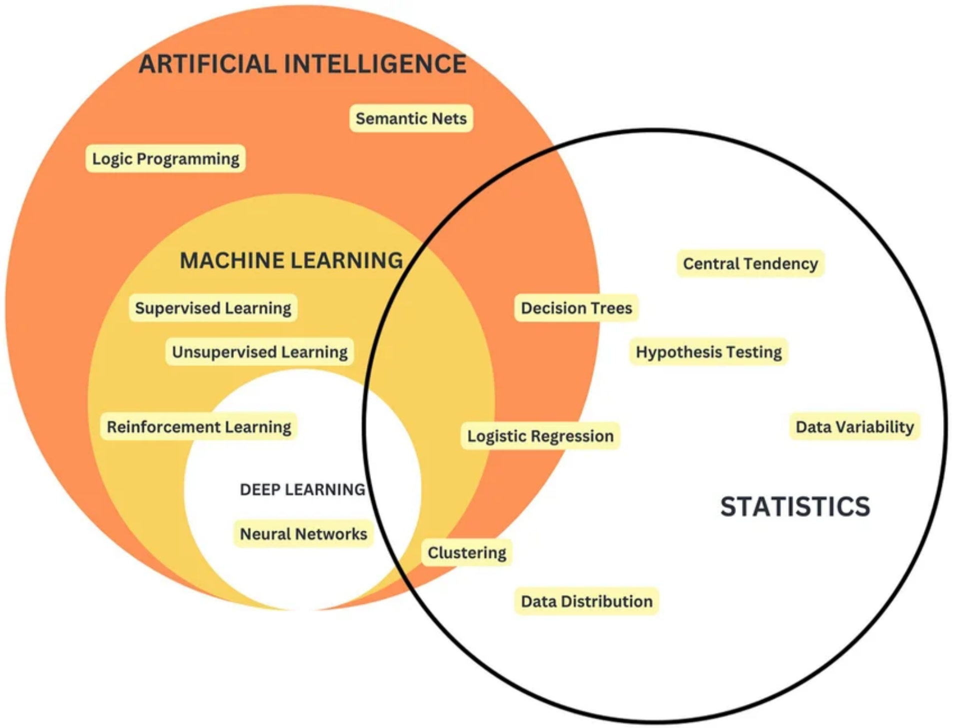
My Goals for The Class

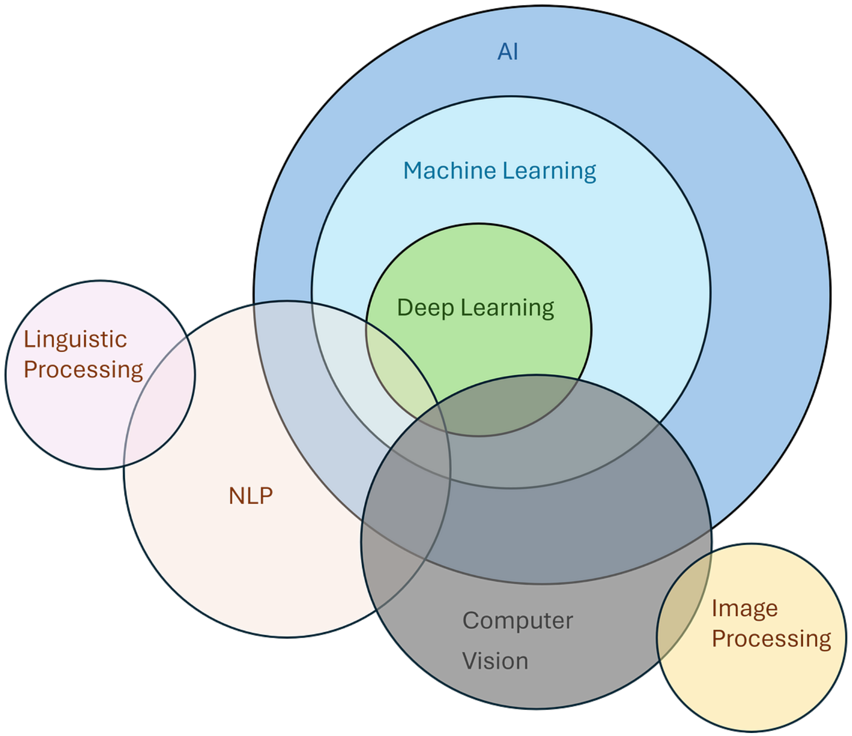
- Systematize you pre-existing knowledge
- Talk about NLP methods and tasks
- Dive deeper into some of the processes behind LLMs
- Give you practical examples of research projects
- Try to answer your questions
- Equip you with resources

Couple of Questions to You

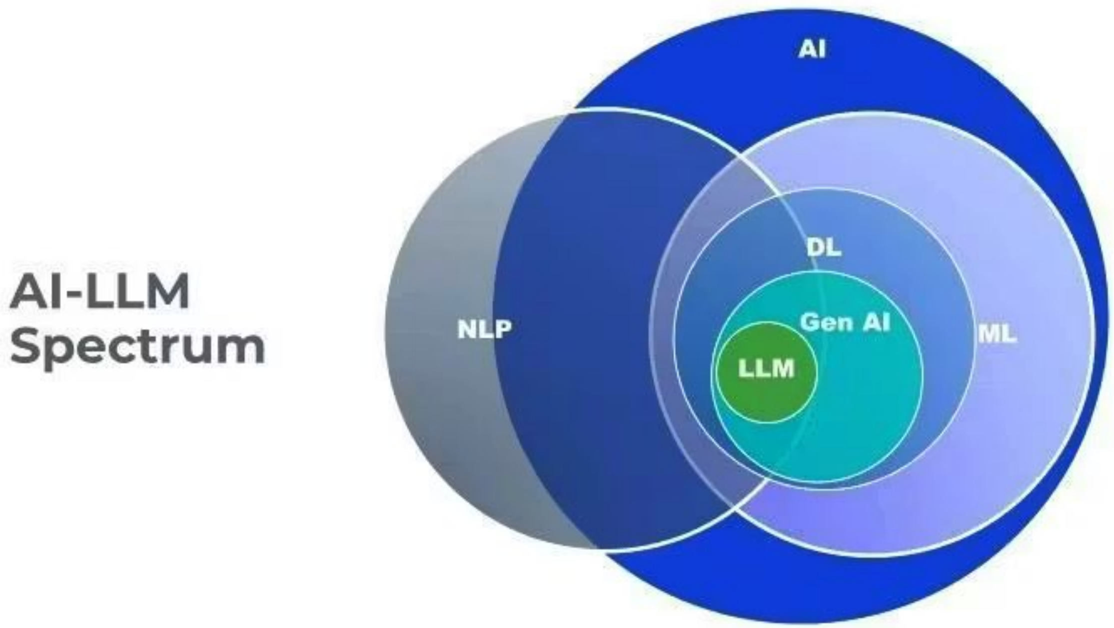
AI Buzzwords

- LM, LLM, AI, ML, DL, NLP, CL, NN, GPT, NWP, CNN, RNN … wtf?



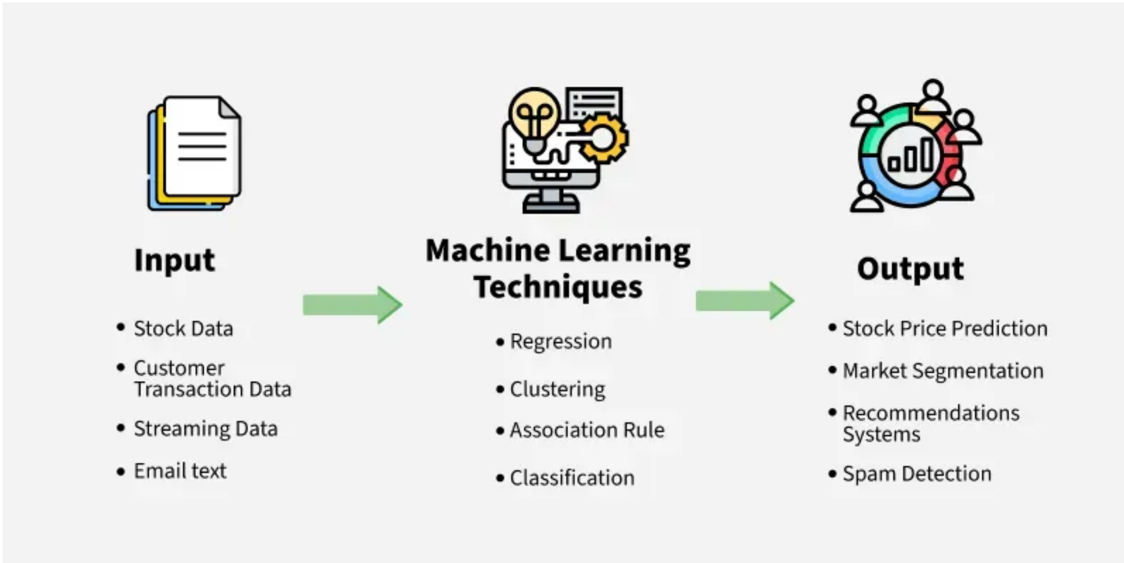


AI Buzzwords

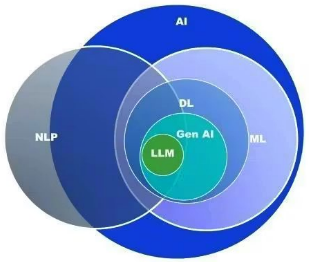


Artificial Intelligence and Machine Learning

- AI: branch of computer science, that deals with **intelligent agents**, systems that can **reason**, **learn**, and **act autonomously**
  - LLMs and chatbots are only one of many applications of AI
- ML: subset of AI, algorithms that can make **predictions based on the data**



AI-LLM Spectrum



Deep Learning and Neural Networks

- DL: type of ML that uses Neural Networks
- NN can process more complex patterns than traditional ML algorithms

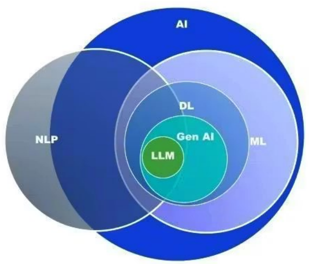
House prices from simple inputs like **size**, **location**, and **number of rooms**:

linear regression

Classifying images of cats and dogs from inputs like **texture**, **shape**, **edges** (complex visual patterns):

NNs

AI-LLM Spectrum



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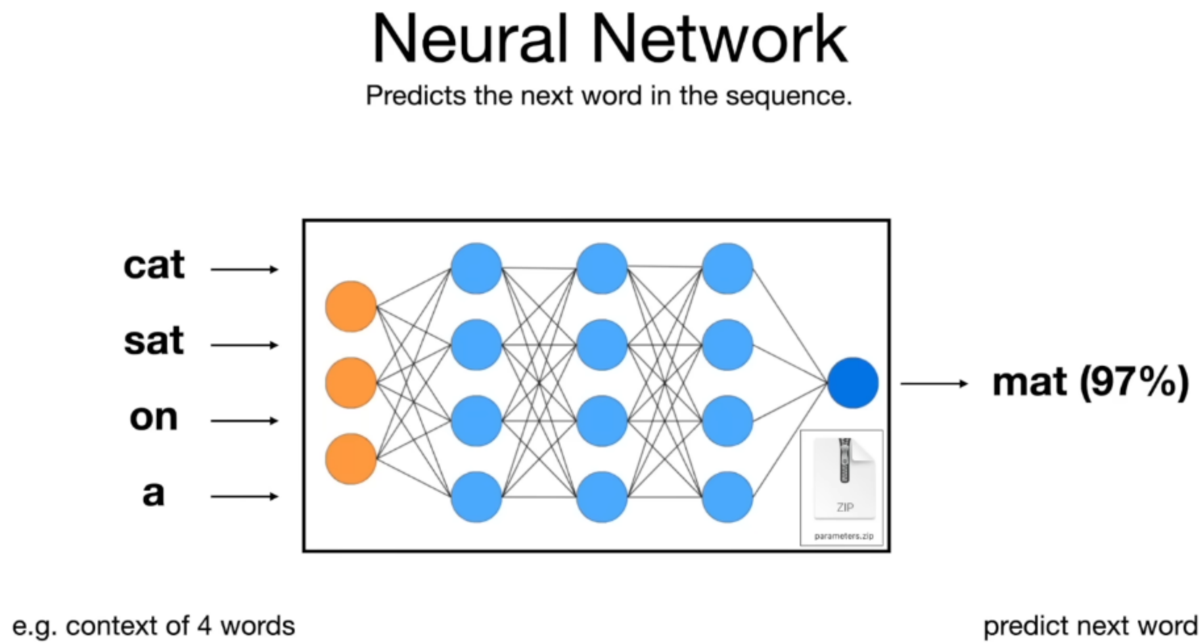
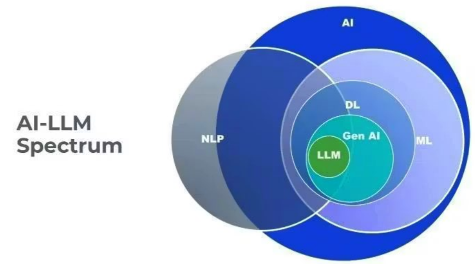
Neural Network

- A **ML algorithm** inspired by the human brain
- Layers of "neurons" that process information
- Learn from data by **adjusting "weights"** between nodes (learning from errors)
- Pattern recognition, predictions, data classification for computer vision, natural language processing, and speech recognition



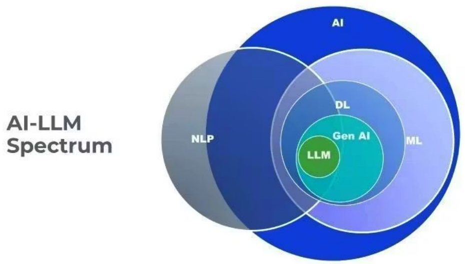


Language Models, Natural Language Processing, Computational Linguistics

- AI, ML, DL, and NNs: how machines **learn**
- NLP and LMs: how machines **work with human language**
- CL?: formal/computational description of languages as a system

Computational Linguistics → NLP → Language Models

language theory → language processing → language generation



Linguistics vs Computational Linguistics

- Lingusitics: the rules followed by languages as a **system**, explains the language rules

The **red** **car**

- CL: **formal** or **computational** description of rules that languages follow, formalizes those rules for computers

Adjectives come before the **nouns**

Generative Pretrained Transformer

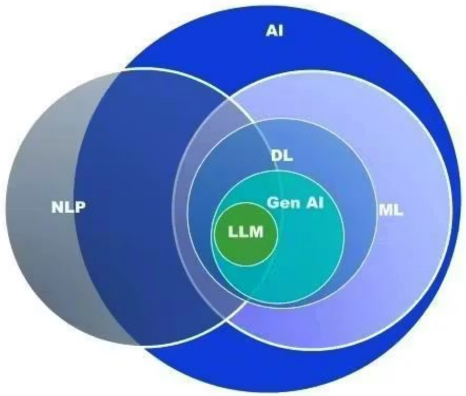
- GPTs are based on a **deep learning architecture** called the **transformer**

Generative → a part of generative AI

Pre-trained → it learned from a lot of existing text before being used

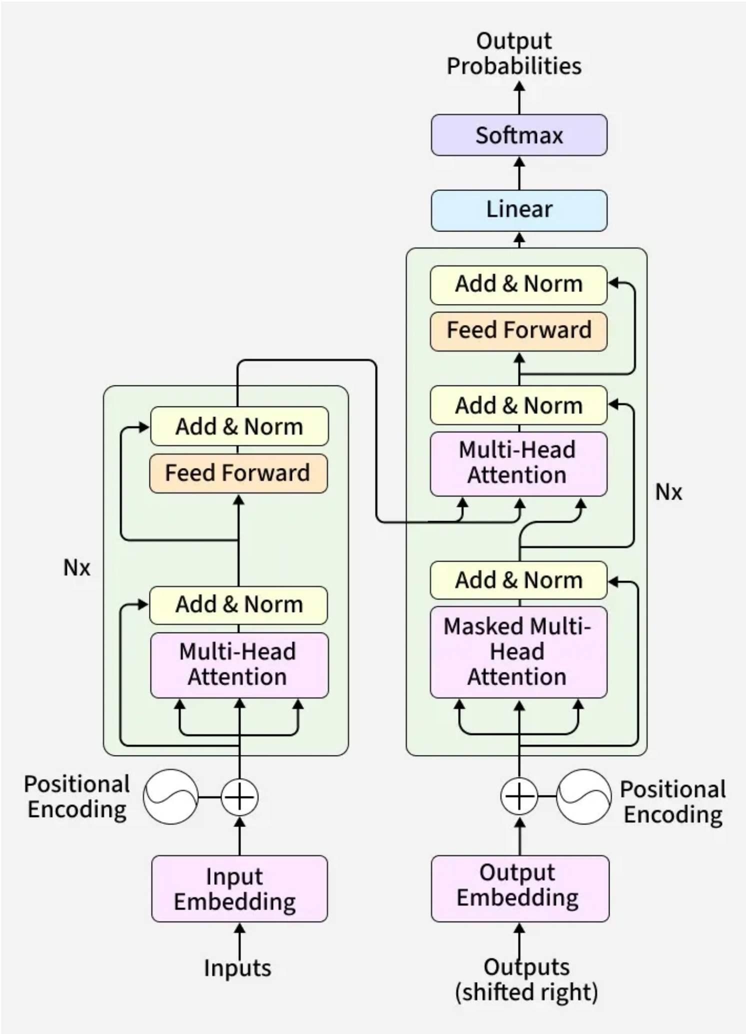
Transformer → the neural network architecture it is based on

AI-LLM Spectrum



Transformer Architecture

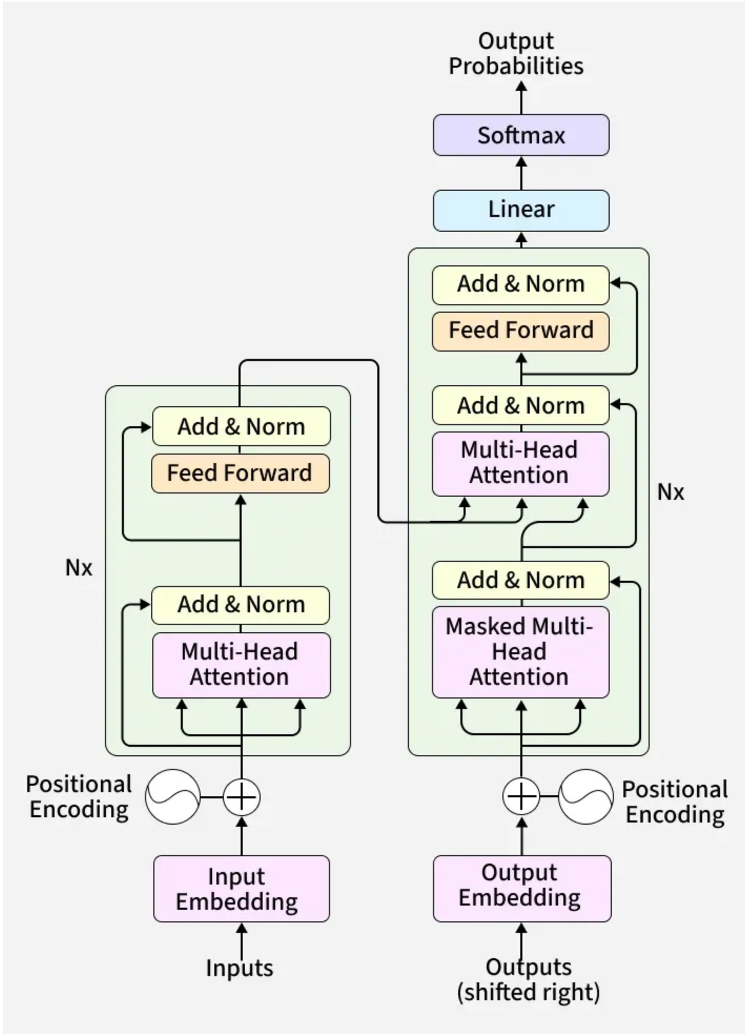
Transformer is a deep learning neural network architecture based on the multi-head attention mechanism, in which text is converted to numerical representations called tokens, and each token is converted into a vector via lookup from a word embedding table. At each layer, each token is then contextualized within the scope of the context window with other (unmasked) tokens via a parallel multi-head attention mechanism, allowing the signal for key tokens to be amplified and less important tokens to be diminished.



Transformer Architecture

- A type of neural network that:
  - understands **relationships between words** in a **sequence all at once** (not one by one)
  - looks at all words **in relation to each other** and focus on the parts that matter most using the **attention heads**

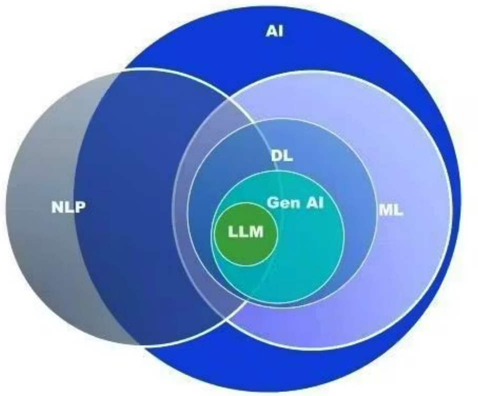
The **animal** didn’t cross the **street** because **it** was too tired.



No Golden Rule

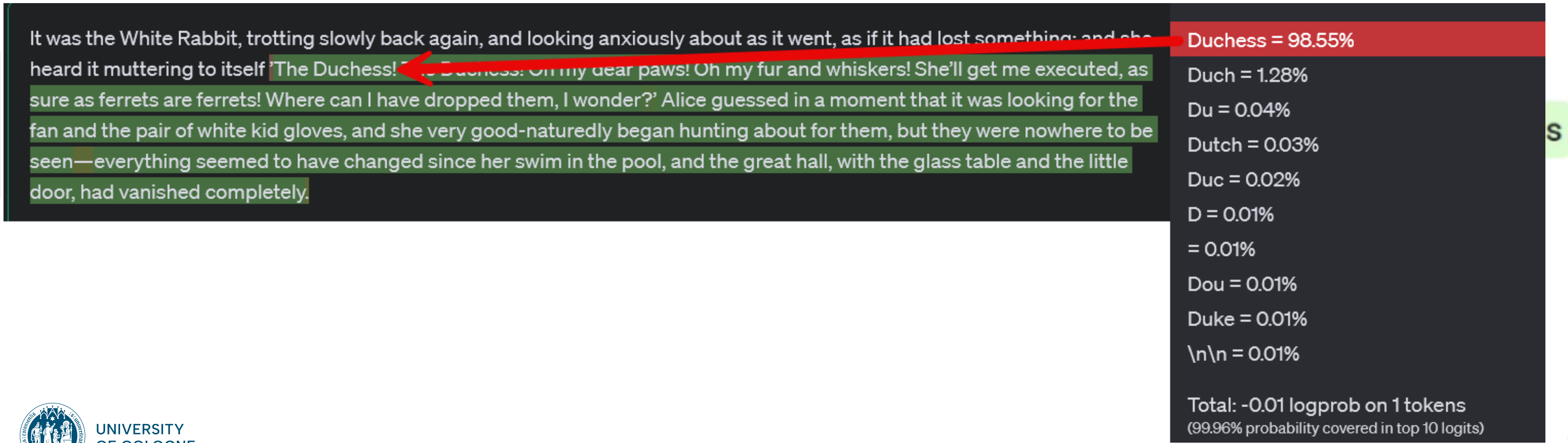
- The choice of the algorithm, model, and methods is always **task-dependent**
- Just because something has a NN or is an LLM, **does not mean** that it will be a **good choice** for your task
- Why choose a computationally-expensive and more complicated method for a simple task?

AI-LLM
Spectrum



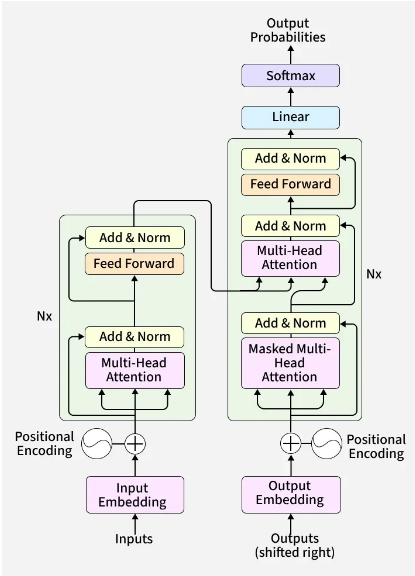
What Do LLMs Actually Do?

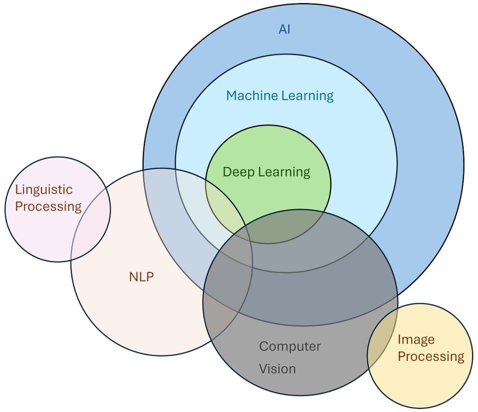
Next (word) token prediction: state-of-the-art language models like GPT predict the next token in a sequence of tokens (**context**) based on their training data. Each predicted token becomes part of the context for the next prediction (**autoregressive**). This simple mechanism, impressive at scale scaled up massively, produces the behaviors we observe.

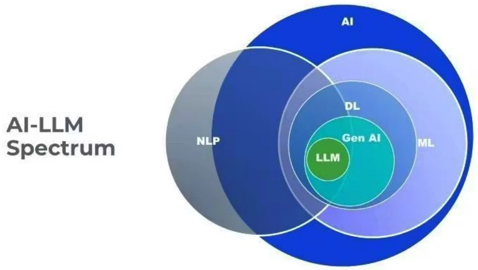


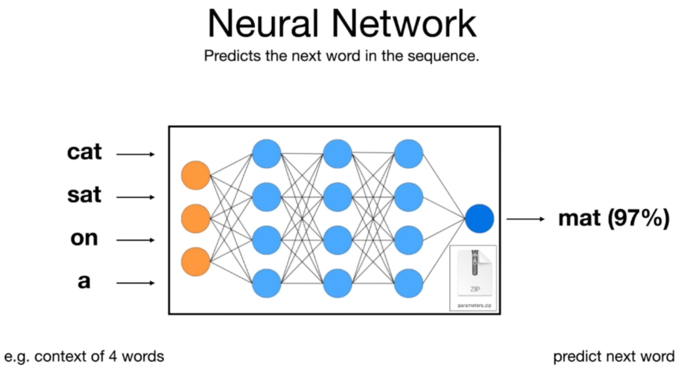
Wrap-Up & Q&A

- Depending on the task we can choose from different ML or DL algorithms
- NLP and LMs: how we can **process and generate language** using different architectures
- Demystifying **transformer** architecture
- **NWP** is the foundation of LMs









Next Up

- Issues, goals, and applications of NLP
- Text preprocessing, text representation, and language modeling
- Language modelling as one of the NLP techniques
- Large Language Models: types, training, and applications
- Practical part: NLP methods in text preprocessing, text representation, and language modeling

Thank you!

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Self Learning Systems Lab

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